# INCIDENT REPORT

SOC Home Lab — AWS Environment

*NIST SP 800-61 | Prepared by: Rashid Lehri*

## 1. INCIDENT SUMMARY

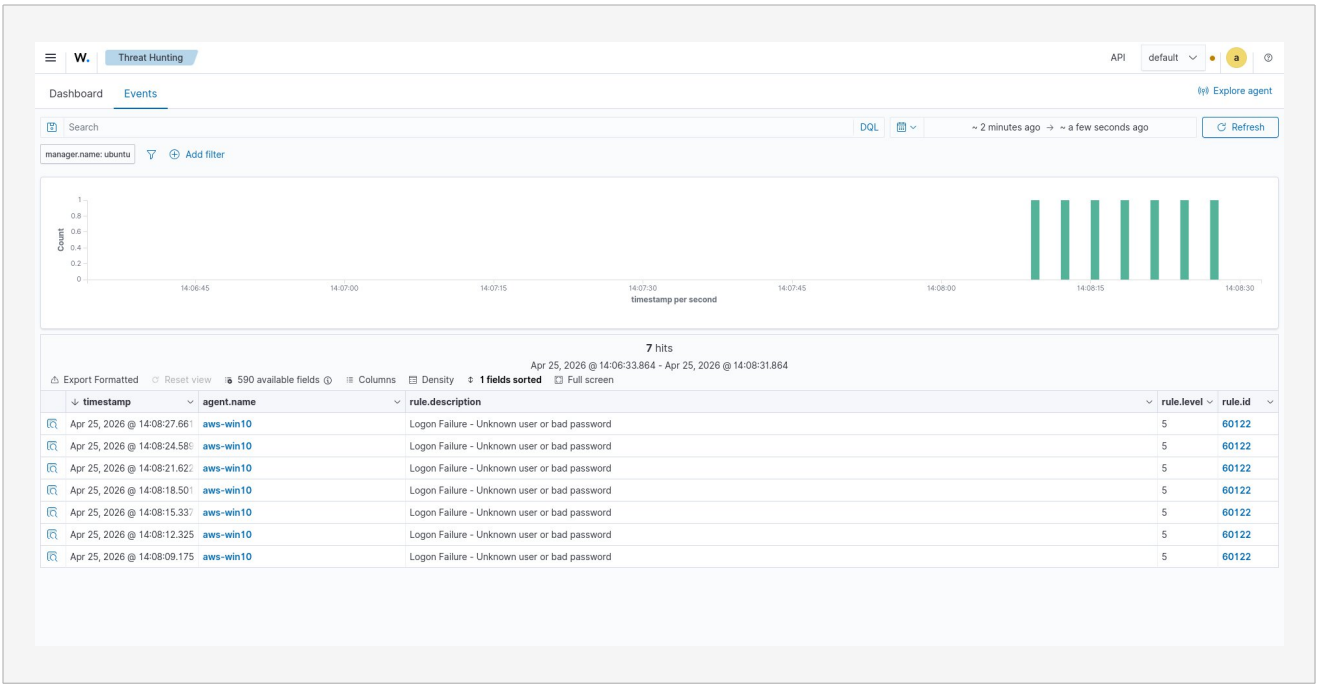
| | |
|---|---|
| **Incident ID** | INC-2024-001 |
| **Date & Time** | 27,april 26  2:00 pm |
| **Prepared By** | Rashid Lehri |
| **Environment** | AWS EC2 — dubaihq.local |
| **Attack Type** | Brute Force Attack via RDP |
| **MITRE Technique** | T1110 — Brute Force |
| **Severity** | HIGH |
| **Final Status** | RESOLVED |
| **Ticket** | osTicket #966712 |

## 2. IDENTIFICATION

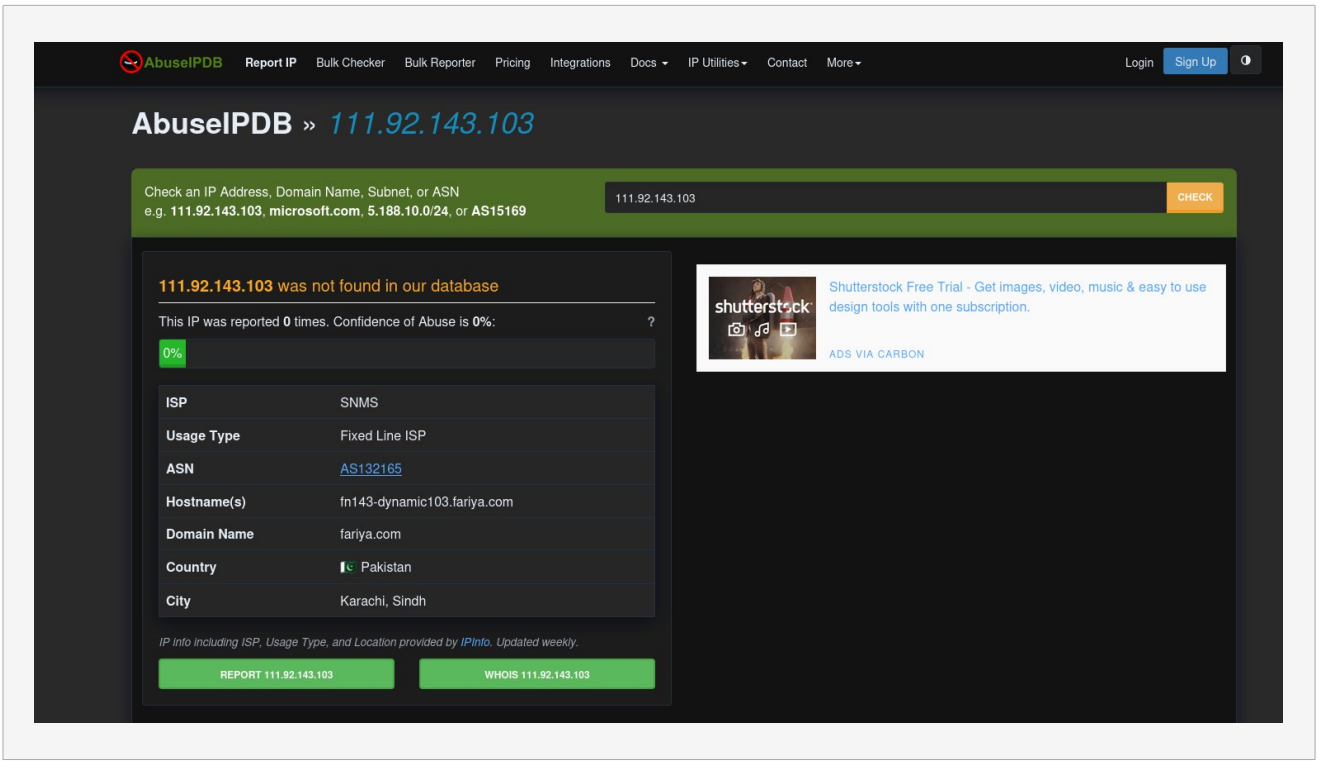
Wazuh SIEM generated a high-severity alert when an external IP repeatedly attempted to access the RDP service on a Windows 10 machine using incorrect passwords. The activity was confirmed through Sysmon logs.

| | |
|---|---|
| **Attacker IP** | 100.111.40.122 |
| **Target Machine** | Windows 10 — dubaihq.local |
| **Target Username** | Employee end-point |
| **Attack Time** | 2:08 PM |
| **Total Attempts** | 7 failed login attempts |
| **Detection Tool** | Wazuh SIEM + Sysmon |
| **AbuseIPDB Score** | . 0% malicious — Test ip |
| **Wazuh Rule ID** | 60122 |

**Screenshot 1 — Wazuh Alert (Brute Force Detected)**




**Screenshot 2 — AbuseIPDB Threat Intel Check**(test ip)




## 3. CONTAINMENT

Immediately after the alert was detected, an automated Python script performed the following actions:

- The Python script automatically detected the brute force alert via the Wazuh API
- The attacker IP's threat score was checked using the AbuseIPDB API
- The attacker IP was blocked by updating the Security Group using AWS Boto3
- Ticket #1042 was automatically created in osTicket (Status: In Progress)
- A Telegram bot sent an instant notification to the security team

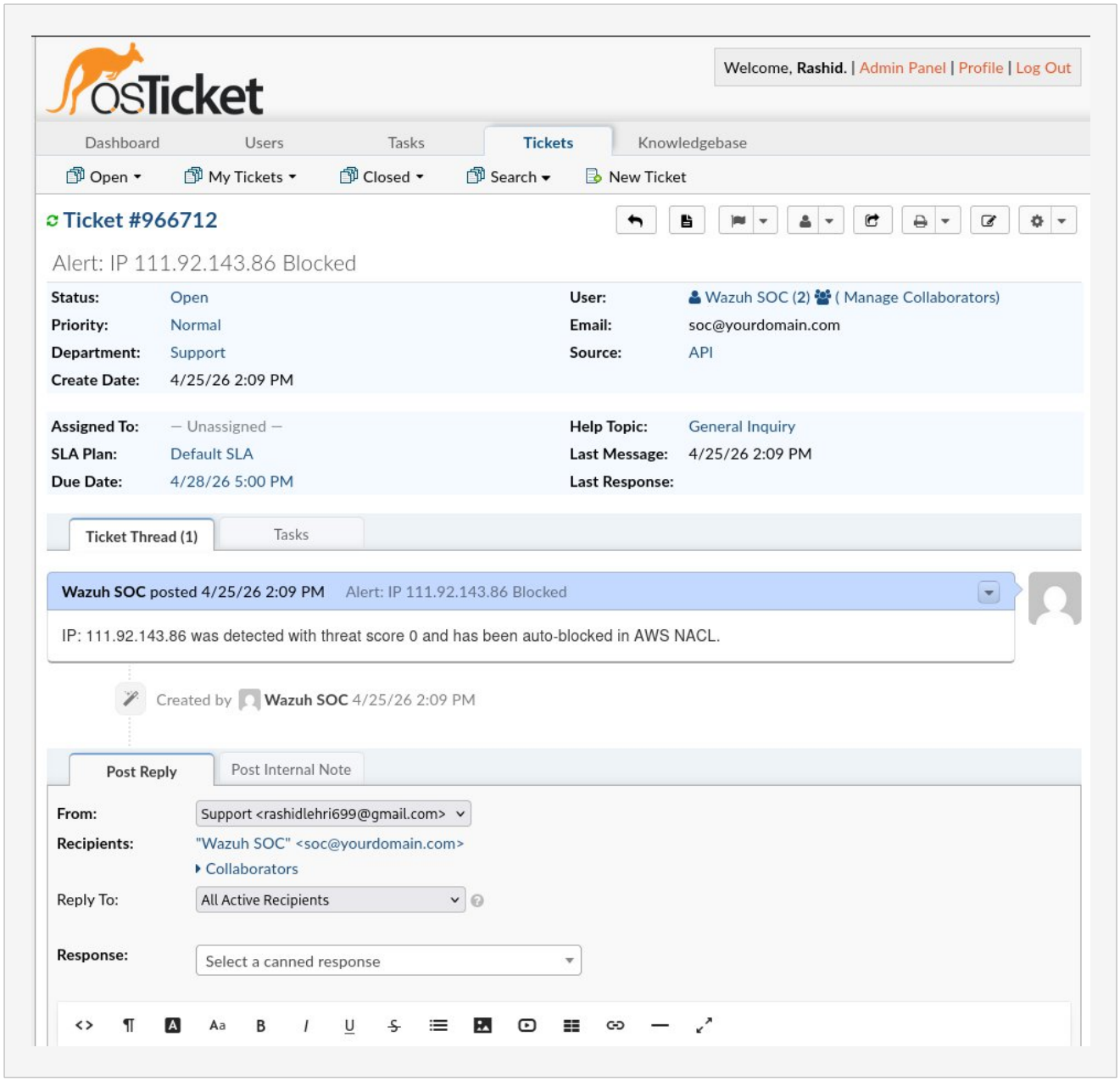
| | |
|---|---|
| **IP Blocked At** | 18:33:06 |
| **Response Time** | ~30 seconds ully automated |
| **Blocked Via** | AWS Security Group — Boto3 |
| **Ticket Created** | osTicket ##868100 |
| **Team Notified** | Telegram Alert |

**Screenshot 3 — Python Script Execution**

```
rashid@ubuntu:~$ python3 soc_auto1.py
── SOC AUTOMATION SCRIPT STARTED ──
[*] Checking Wazuh for Brute Force Alerts in the last 5 minutes ...
[sudo] password for rashid:
[+] ALERT FOUND! MITRE T1110 Detected. Rule: 60122
[+] Attacker IP Extracted: 111.92.143.86
[*] Checking IP: 111.92.143.86 on AbuseIPDB ...
[+] Result: Threat Score for 111.92.143.86 is 0%
[!] DANGER: High Risk IP detected! Initiating Auto-Remediation ...
[*] Boto3 Activated: Blocking 111.92.143.86 in AWS NACL ...
[+] SUCCESS: IP successfully blocked in AWS Firewall!
[*] Creating automated Incident Ticket in osTicket ...
[+] SUCCESS: Ticket #868100 created automatically!
[*] Sending Alert to Telegram ...
[+] SUCCESS: Telegram Notification Sent!

[🏁] FULL AUTOMATION CYCLE COMPLETED SUCCESSFULLY!
```

**Screenshot 4 — osTicket #1042 Created**




**Screenshot 5 — Telegram Notification**

12:41 AM

Rashid_soc_alerts
Connecting...

Set Profile Photo

April 24

/start 11:57 AM

Hello 11:58 AM

Hi 11:59 AM

Hi 11:59 AM

How 12:01 PM

Hi 12:03 PM

🚨 **CRITICAL SOC ALERT** 🚨

🔥 **Attack:** Brute Force Detected
🌐 **Attacker IP:** 185.156.73.125
📊 **Abuse Score:** 100%
🛡️ **Action:** Blocked via AWS NACL
📄 **osTicket:** #595305

✅ **Status:** Remediation Complete. 12:13 PM

April 25

🚨 Critical Alert: MITRE T1110 Detected
Attacker IP: 111.92.143.86
AbuseIPDB Score: 0% malicious
Action: IP Blocked via AWS
Ticket: #Auto
Created Time: 18:33:06 11:33 PM

Message

## 4. ERADICATION

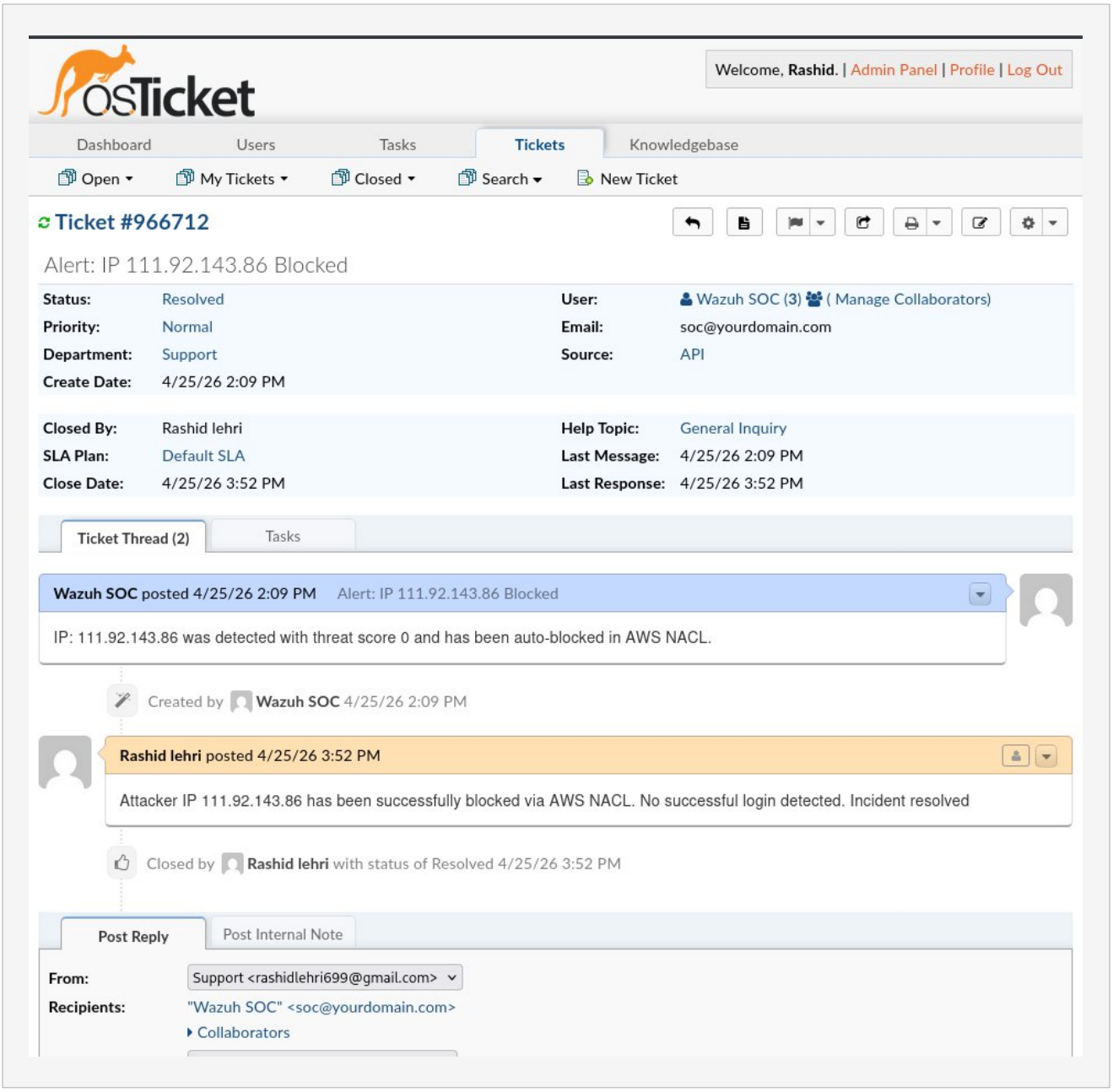
- The attacker IP was permanently blocked in the AWS Security Group
- Wazuh false positive rules were tuned — 1–2 attempts are ignored, while 10+ attempts trigger a HIGH alert
- RDP public access was restricted — only whitelisted IPs are allowed
- Windows Event Logs on the target machine were reviewed — no successful login was found
- Sysmon logs were verified — no malicious processes or persistence mechanisms were detected

## 5. RECOVERY

The systems were restored to normal operation. No data breach or unauthorized access was confirmed.

| | |
|---|---|
| **System Status** | CLEAN — No compromise found |
| **Accounts Checked** | All AD accounts verified — No active suspicious sessions |
| **Logs Verified** | Windows Event Logs + Wazuh Logs |
| **Recovered At** | 3:52 PM |
| **Ticket Status** | osTicket #1042 — RESOLVED |

**Screenshot 6 — osTicket Resolved**




## 6. LESSONS LEARNED

| What Worked Well | Improvements Made |
|---|---|
| • Wazuh detected the attack in real time<br>• The Python automation responded within 30 seconds<br>• AbuseIPDB confirmed the IP's malicious history | • Wazuh false positive rules were tuned<br>• AbuseIPDB threat intelligence was integrated<br>• The T1110 Brute Force playbook was documented<br>• RDP access controls were strengthened |

| | |
|---|---|
| • The osTicket workflow functioned smoothly<br>• The Telegram alert was received instantly | • The full project was documented on GitHub |

## 7. REPORT DETAILS

| | |
|---|---|
| **Analyst Name** | Rashid Lehri |
| **Role** | SOC Analyst — Home Lab Project |
| **Date Prepared** | 27,april 26 |
| **Report Version** | v1.0 — Final |